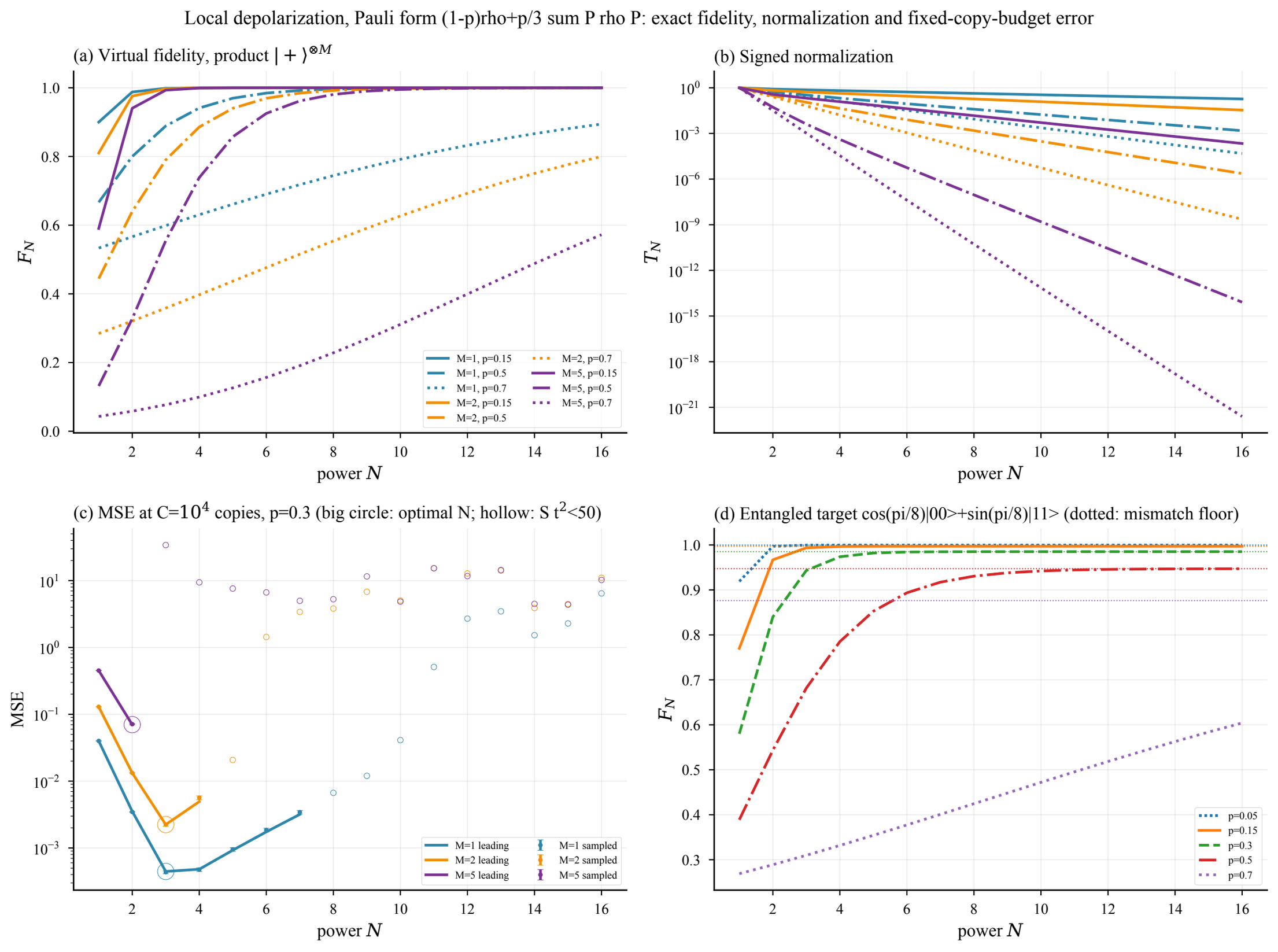
Local depolarization, Pauli form (1-p)rho+p/3 sum P rho P: exact fidelity, normalization and fixed-copy-budget error
(a) Virtual fidelity, product $|+\rangle^{\otimes M}$
F_N
power N
M=1, p=0.15
M=1, p=0.5
M=1, p=0.7
M=2, p=0.15
M=2, p=0.5
M=2, p=0.7
M=5, p=0.15
M=5, p=0.5
M=5, p=0.7
(b) Signed normalization
T_N
power N
(c) MSE at C=10^4 copies, p=0.3 (big circle: optimal N; hollow: S t^2<50)
MSE
power N
M=1 leading
M=2 leading
M=5 leading
M=1 sampled
M=2 sampled
M=5 sampled
(d) Entangled target cos(pi/8)|00>+sin(pi/8)|11> (dotted: mismatch floor)
F_N
power N
p=0.05
p=0.15
p=0.3
p=0.5
p=0.7